EchoSense

Voice- Assisted Patient Monitoring

Introduction

- **The Communication Challenge**
- Effective communication is vital for expressing needs, emotions, and emergencies.
- Patients with severe motor disabilities or speech disorders often struggle to communicate.
- This difficulty creates significant barriers in patient-caregiver interactions.

- **EchoSense: An Intelligent Solution**
- Recent AI and speech recognition advancements enable new assistive communication systems.
- These technologies can help patients communicate their needs more efficiently.
- EchoSense offers a simple, intelligent solution to enhance patient-caretaker communication.

PROBLEM STATEMENT

Limitations of Current Methods

Paralysed patients with speech impairments, such as Dysarthria, struggle to communicate their needs clearly and effectively.

Critical Impact

Traditional communication methods are often slow, inconvenient, or require physical interaction, posing significant barriers.

PROPOSED SOLUTION

What is EchoSense?

An AI-based assistive communication system. Designed to help patients efficiently communicate their needs to caregivers.

How It Works

Employs Artificial Intelligence and speech recognition technology. Detects patient speech, identifies critical needs, and generates timely alerts for caregivers.

WORKFLOW

Patient speaks
↓
Voice Activity Detection detects speech
↓
Audio is recorded through microphone
↓
Speech recognition converts speech → text
↓
Keyword detection identifies critical needs
↓
Alert generated for caretaker
↓
Caretaker receives message and audio notification

Advantages

Key Advantages

Resource Lightweight : Designed for efficiency, consuming minimal system resources.

No Advanced AI Dependency : Avoids the need for heavy, complex artificial intelligence processing.

Intuitive User Interface : Offers an easy-to-navigate and accessible user experience.

RESULTS AND OUTCOMES

- **Key Achievements**

Voice-based keyword detection works

Alerts reach caregiver dashboard

Medication tracking works

Patient history is recorded

Communication between patient and caregiver is improved

FUTURE SCOPE

Training models using TORGO Dataset to improve recognition of dysarthric speech

Improved noise reduction and speech accuracy

Integration with hospital monitoring systems

Multi-patient monitoring

CONCLUSION

EchoSense demonstrates how AI-based speech recognition can improve communication between patients and caregivers.

Overall, EchoSense provides a practical and scalable approach to supporting elderly or vulnerable patients by enabling faster communication and better monitoring through intelligent voice detection

Thank You